Solution:

Here STP means standard temperature (T=273K or 0°C) and Pressure (P=1 atm or 101.3 kPa)

We can use ideal gas equation $V = \frac{\mu RT}{P}$.

Here μ = 1 mol and R =8.314 J/mol.K.

By substituting the values

$$V = \frac{(1\ \text{mol})(8.314\ \frac{\text{J}}{\text{mol}}\ \text{K})(273\text{K})}{1.013\times10^{5}\ \text{Nm}^{-2}}$$

$=22.4 \times 10^{-3}\ \text{m}^3$

We know that 1 Litre (L) = $=10^{-3}\text{m}^3$. So we can conclude that 1 mole of any ideal gas has volume 22.4 L.

By multiplying 22.4L by $\frac{300K}{273K}$ we get the volume of one mole of gas at room temperature. It is 24.6 L.

EXAMPLE 8.5

Estimate the mass of air in your class room at NTP. Here NTP implies normal temperature (room temperature) and 1 atmospheric pressure.



Solution

The average size of a class is 6m length, 5 m breadth and 4 m height. The volume of the room V = 6 × 5 × 4 = 120m^3. We can determine the number of mole. At room temperature 300K, the volume of a gas occupied by any gas is equal to 24.6L.

The number of mole $\mu = \frac{120\text{m}^3}{24.6\times10^{-3}\text{m}^3}$ ≈4878 mol.

Air is the mixture of about 20% oxygen, 79% nitrogen and remaining one percent are argon, hydrogen, helium, and xenon. The molar mass of air is 29 gmol^{-1}.

So the total mass of air in the room $m = 4878 \times 29 \times 10^{-3} = 141.4\text{kg}$.

8.2.2 Heat capacity and specific heat capacity

Take equal amount of water and oil at temperature 27°C and heat both of them till they reach the temperature 50°C. Note down the time taken by the water and oil to reach the temperature 50°C. Obviously these times are not same. We can see that water takes more time to reach 50°C than oil. It implies that water requires more heat energy to raise its temperature than oil. Now take twice the amount of water at 27°C and heat it up to 50°C , note the time taken for this rise in temperature. The time taken by the water is now twice compared to the previous case.

We can define 'heat capacity' as the amount of heat energy required to raise the temperature of the given body from T to T + ΔT .

$$\text{Heat capacity S} = \frac{\Delta Q}{\Delta T}$$

Specific heat capacity of a substance is defined as the amount of heat energy required to raise the temperature of 1kg of a substance by 1 Kelvin or 1°C

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